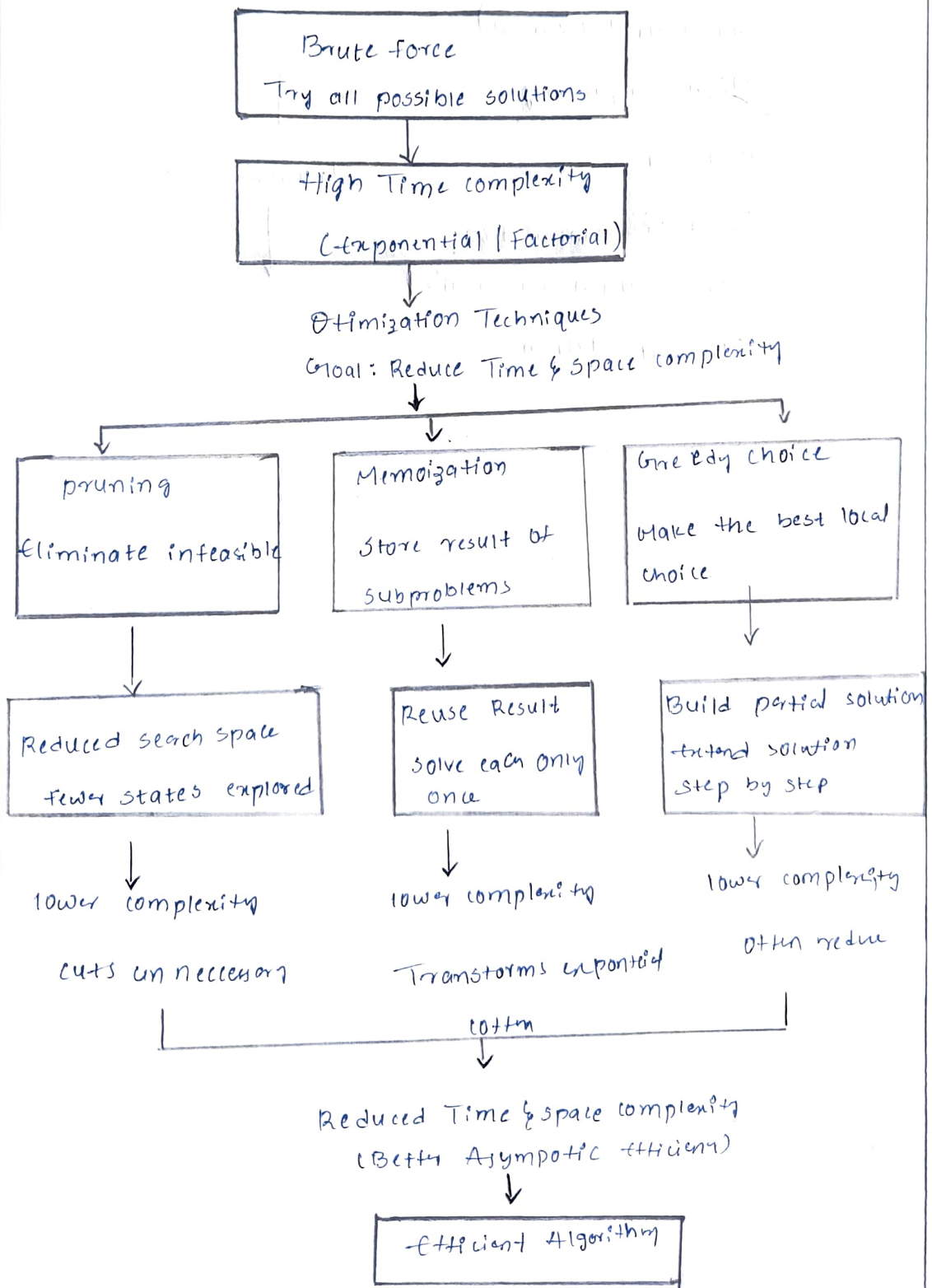


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ans:-

BRUTE FORCE - $\rightarrow$ HIGH COMPLEXITY - $\rightarrow$ OPTIMIZATION TECHNIQUES
 $\rightarrow$ EFFICIENT ALGORITHM



Legend (Complexity Improvement)

- Exponential ($2^n, n!$)

- High complexity

Reduced search

- Avoid Re-computation

polynomial ($n, n \log n, n^2$)

- Efficient solution

Improving efficiency



Multiple Algorithms $\rightarrow$ Comparison Criteria $\rightarrow$ Growth Functions
 $\rightarrow$ Best choice

Multiple Algorithm (Candidate set)
-ex:- Brute force, Divide & Conquer

Comparison Criteria
Evaluate each algorithm using key factor

Asymptotic Analysis

Input Characteristics

Constraints (practical factors)

How does the algorithm scale with input size

- Time complexity
- Space complexity
- Growth function (Big-O)

What is the nature of input data

- Input size (n)
- Data Distribution
- Sorted/Unsorted
- Spars/Dense
- Static/Dynamic

Time limit
Memory limit
Hardware/Platform

Growth Functions (Order efficiency)

$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n n)$
 $< O(n!)$
Worse (less efficient)

Flow Multiple Algorithm Converse

↓
1) Multiple Confident Algorithm

↓
2) compare using all Criteria

↓
3) Rank by growth function

↓
4) Apply All constraint / I/p factor

↓
5) find Decision -> Best Logic